

Intergenerational Housing-Fertility Calibration Note

June 23, 2026. Diagnostic status, not a production calibration.

Active object. The current quantitative object is the one-market, no-location, Markov-income intergenerational model in `code/model/intergen_housing_fertility/`. The old center-periphery `dt_cp_model` code is not the active calibration object for this note. The relevant recent evidence is the full global room-gap search and the 14-moment rescore, both documented under `docs/model/`.

Parameter vector. The search varies 13 parameters:

$$\theta = \{\beta, \alpha, b_0, \bar{c}_0, \bar{c}_n, \bar{h}_0, \bar{h}_{\text{jump}}, \bar{h}_n, \psi_{\text{child}}, \kappa_n, \chi, \theta_0, \theta_n\}.$$

Here β controls saving and old wealth, α the consumption-space tradeoff, b_0 entry wealth/access, $(\bar{c}_0, \bar{c}_n)$ child consumption needs, $(\bar{h}_0, \bar{h}_{\text{jump}}, \bar{h}_n)$ housing needs, ψ_{child} and κ_n fertility level/dispersion, χ the owner-service premium, and (θ_0, θ_n) bequest level and fertility scaling.

Preferred diagnostic objective. The previous full room-gap target system had 17 hard moments for 13 parameters. For interpretation, the cleaner diagnostic objective is 14 hard moments: one more moment than parameters. The three demoted moments remain diagnostics, not discarded information.

Moment	Target	Model	Gap	Weight	Role
TFR	1.700	1.764	0.064	20.0	Fertility level
Childless rate	0.150	0.276	0.126	20.0	Fertility extensive margin
Overall ownership	0.575	0.490	-0.085	100.0	Tenure level
Parent-child ownership gap	0.168	0.291	0.123	45.0	Family tenure wedge
Housing increment 0 → 1 child	0.664	1.110	0.446	14.0	First-child housing demand
Housing increment 1 → 2 children	0.488	0.331	-0.157	8.0	Marginal child housing demand
Young liquid wealth/income	0.179	0.593	0.414	12.0	Access and saving
Old nonhousing wealth/income median	2.230	2.059	-0.172	0.8	Lifecycle saving
Old parent-childless NH wealth gap	1.007	0.291	-0.717	2.0	Bequest/composition
Childless renter mean rooms	3.805	4.560	0.755	6.0	Renter space level
Owner-renter room gap, childless	2.419	1.493	-0.926	12.0	Space separation
Childless owner share rooms ≥ 6	0.596	0.762	0.165	25.0	Owner large-home use
Old ownership	0.764	0.877	0.113	160.0	Old retention/exit
Young ownership, ages 25–34	0.341	0.175	-0.166	80.0	Entry/access
<i>Diagnostics, not hard in the 14-moment rescore</i>					
Childless owner mean rooms	6.224	6.0	6.053	-0.171	Implied by renter mean plus gap
Childless renter share rooms ≥ 6	0.138	25.0	0.100	-0.038	Renter large-unit tail
Old parent-childless ownership gap	0.083	40.0	0.146	0.063	Old family tenure diagnostic

Current best fit. The best retained record under both the 17-moment loss and the 14-moment rescore is DE wave 4, task 3, case 492. Its 14-moment loss is 26.530; its original 17-moment loss is 26.903. Dropping the three diagnostic moments lowers the loss level slightly but does not change the selected basin.

Identification map. Fertility level and childlessness discipline κ_n , ψ_{child} , and child costs. The two child-housing increments discipline $\bar{h}_{\text{jump}}$ and $\bar{h}_n$. Overall ownership, young ownership, parent-child ownership gaps, and young liquid wealth discipline entry/access and tenure parameters, especially b_0, χ, α, β . Renter rooms, the owner-renter room gap, and owner large-home use discipline the space-demand and owner-ladder block. Old ownership and old wealth moments discipline old retention, saving, and bequests, especially β, θ_0 , and θ_n . This is why removing a hard moment requires a replacement moment for the same block.

Main tension. The model does not need old households to exit ownership in an absolute sense. The target is old ownership 0.764; the best model gives 0.877. At the same time, young ownership should be 0.341, but the model gives 0.175, and the owner-renter room gap should be 2.419, but the model gives 1.493. The hard problem is joint: young entry is too low, old retention is too high, and owner-renter space separation is too small.

The all-case frontier is sharper than the scalar loss. Among 263,282 retained records, zero cases satisfy young ownership ≥ 0.20 , old ownership ≤ 0.90 , and room gap ≥ 1.50 . Cases with old ownership and room gap close to target usually have young ownership near zero. Cases with young ownership close to target usually keep old ownership near one and compress the room gap. This suggests the current model lacks a separating mechanism between young entry, old retention, and large owner space.

Next work. The 14-moment objective is cleaner and should be made explicit if it becomes the diagnostic benchmark. But rescoring the existing search shows that this alone does not fix the frontier. The next useful work is a focused verification of the main frontier records, followed by a mechanism audit of tenure/access, starter ownership versus large-owner housing, old downsizing/retention, and old nonhousing wealth composition. Another blind global search is lower value unless it tests a specific mechanism or an explicitly revised target system.